MAJOR PROJECT PRESENTATION • COMPUTER SCIENCE & ENGINEERING

Context-Aware SDN

Unified Multi-Class Network Anomaly Classification and Differentiated Auto-Remediation in Software-Defined Networks

SDN ARCHITECTURE
OpenFlow 1.3 & Ryu Controller

5 ANOMALY CLASSES
Normal / Flash / DDoS / Congestion / Fail

ENSEMBLE ML
Random Forest & Gradient Boosting

SURGICAL MITIGATION
Load Balancing, Reroute, Ingress Filter

INTRODUCTION & MOTIVATION

The Core Challenge: Blind, One-Size-Fits-All Reaction in SDNs

Why modern cloud & enterprise networks suffer massive collateral damage during disruptions.

The 'One-Size-Fits-All' Flaw

- Conflation of Surges with Attacks: Traditional intrusion defense systems treat legitimate traffic spikes (e-commerce flash sales) identically to malicious DDoS floods, dropping 60–80% of revenue-generating client requests.
- Reactive Route Flapping: Gradual buffer queue buildup triggers abrupt link resets or uncoordinated rerouting, creating severe oscillations across core paths.
- Siloed Management Planes: Physical infrastructure health (port-down, fiber cuts) and packet-level anomaly detection operate in disconnected subsystems without shared telemetry.
- Catastrophic Collateral Loss: Coarse mitigation policies punish benign users, directly violating Service Level Agreements (SLAs).

The Context-Aware SDN Solution

✓ Root-Cause Anomaly Discrimination: Accurately categorizes network state into 5 discrete conditions: Normal, Flash-Crowd Surge, DDoS Flood, Slow Congestion, and Link Failure.

✓ Fused Cross-Layer Telemetry: Combines rolling-window flow statistics (Shannon entropy, packet rates, size variance) with direct OpenFlow 1.3 asynchronous port status signals.

✓ Class-Specific Surgical Remediation: Replaces generic drop rules with tailored OpenFlow actions: Dynamic Multipath Load Balancing, Ingress Flow Drops, Preemptive Rerouting, and Sub-50ms Failover.

✓ Zero-Collateral-Damage Guarantee: Neutralizes cyber-threats at the perimeter while preserving 100% of legitimate client throughput.

LITERATURE REVIEW

Analysis of Closely Related Works in SDN Anomaly Detection & Mitigation

Evaluation of four seminal state-of-the-art papers representing key aspects of the problem space.

Paper 1: DDoS Detection via ML in SDN

Alanazi et al. (MDPI Telecom, 2025)

- Focus: ML-based volumetric attack detection using Random Forest & KNN in Mininet/Ryu.
- Telemetry: Polled flow tables for packet/byte counters and flow duration.
- Key Contribution: Demonstrated high binary detection accuracy (>98%) on static DDoS attack datasets.

Paper 2: DDoS vs. Flash Crowd Discrimination

Sahoo, Puthal, Tiwary et al. (Springer, J. Supercomput., 2021)

- Focus: Differentiating malicious DDoS attacks from legitimate flash crowds using information entropy.
- Telemetry: Calculated Shannon entropy on IP sources and packet arrival distributions.
- Key Contribution: Proved that high entropy reflects flash crowds while low entropy indicates DDoS.

Paper 3: Link Fault Recovery & Congestion

Zhang, Wang, & Zhao (IEEE Access / ACM, 2023)

- Focus: Dynamic hierarchical single link fault recovery and congestion restoration in SDNs.
- Telemetry: Switch port status events and path queue utilization monitoring.
- Key Contribution: Devised dynamic backup path computation to relieve bottleneck link saturation.

Paper 4: Multi-Class Attack Identification

Alrashdi, Alqazzaz, & Tariq (arXiv / IEEE, 2024)

- Focus: Multi-class DDoS identification (UDP flood, SYN flood, ICMP flood) using meta-learning ensembles.
- Telemetry: Multi-controller placement and rolling flow statistics.
- Key Contribution: Achieved fine-grained classification among distinct attack vectors in large topologies.

RESEARCH GAP

Critical Limitations of Existing Literature & The Unaddressed Gap

Direct comparative analysis of the 4 reference papers highlighting their architectural deficiencies.

Paper 1: Alanazi et al. (2025)

Limitation: Binary-Only Classification

- Inability to discriminate traffic semantics: Strictly binary (DDoS vs Normal).
- Severe False Positives: Treats all high-volume spikes as attacks, causing massive collateral drops during flash-crowd sales.
- No differentiated action: Employs generic all-or-nothing flow drop rules.

Paper 2: Sahoo et al. (2021)

Limitation: Isolated Traffic-Only Scope

- Ignores Physical Layer Faults: Completely blind to link breaks, fiber cuts, and port failures.
- Omits Congestion Buildup: Does not detect or handle progressive queue buffer exhaustion.
- Single-Action Mitigation: Lacks dynamic multi-path group table load balancing for flash crowds.

Paper 3: Zhang et al. (2023)

Limitation: Hardware-Only Perspective

- Blind to Cyber-Attacks: Focuses exclusively on physical link outages; cannot detect DDoS or volumetric floods.
- Threshold-Based Heuristics: Uses rigid static thresholds rather than predictive ML anomaly discrimination.
- No Cross-Layer Fusion: Fails to integrate packet-level entropy with topology recovery.

Paper 4: Alrashdi et al. (2024)

Limitation: Lack of Closed-Loop Remediation

- Stops at Classification: Evaluates only offline F1-scores without implementing live OpenFlow mitigation.
- Security-Centric Silo: Completely overlooks operational network degradation (congestion & link failure).
- No Baseline Benchmark: Fails to quantify the QoS benefit of differentiated vs. naive defense.

THE SYNTHESIS GAP: No existing framework unifies traffic-based cyber anomalies (DDoS/Flash Crowds) and infrastructure faults (Congestion/Link Cuts) into a single telemetry-driven classifier with surgical, class-specific OpenFlow auto-remediation.

PROBLEM STATEMENT

Formal Problem Formulation & Core Research Questions

Defining the fundamental challenges in autonomic multi-anomaly SDN management.

1. Semantic Conflation Under Volumetric Bursts

Traditional SDN detection models rely heavily on gross byte and packet rates. Because both DDoS attacks and legitimate flash-crowd surges exhibit abrupt volumetric surges, conventional binary classifiers conflate the two states. Consequently, standard rate-limiting drops legitimate user traffic, inflicting severe collateral damage and financial loss during critical peak-demand periods.

2. Cross-Layer Telemetry Disconnect

Current SDN controllers maintain segregated telemetry pipelines: flow counters are polled for security inspection, while physical port events are processed independently for route updates. The lack of a unified feature-engineering window blinds the controller to inter-dependent cascading failures, such as when a severed core link triggers buffer congestion across secondary backup paths.

3. Destructive Collateral Impact of Generic Mitigation

When anomalies are detected, existing SDN controllers invoke coarse, one-size-fits-all responses—either applying blanket IP subnet drops or initiating blind global route resets. These generic reactions poison alternate paths, induce route flapping, increase average packet latency by over 300%, and degrade overall network QoS.

CORE RESEARCH QUESTION: How can an SDN controller autonomously discriminate between five operational conditions in real time and execute differentiated OpenFlow actions

OBJECTIVES

Primary & Specific Research & Implementation Objectives

A structured engineering and scientific roadmap for the Context-Aware SDN framework.

PRIMARY OBJECTIVE

To design, implement, and empirically validate an autonomic closed-loop Software-Defined Networking architecture capable of real-time multi-class anomaly classification and surgical, differentiated OpenFlow auto-remediation.

Objective 1: Telemetry & Feature Engineering

- Develop an asynchronous Ryu OpenFlow 1.3 monitoring engine.
- Poll flow statistics and port counters at 2.0s rolling intervals.
- Compute a 10D feature vector: Shannon entropy of source IPs, packet size variance, packet/byte rates, port status, and drop counters.

Objective 2: Multi-Class ML Engine

- Build and optimize ensemble classifiers (Random Forest & Gradient Boosting).
- Achieve >95% multi-class accuracy across 5 distinct operational conditions.
- Maintain sub-15ms inference latency to prevent controller processing bottlenecks.

Objective 3: Differentiated Action Engine

- Map classified anomalies to specific OpenFlow 1.3 datapath directives:
  - Flash Crowd: Group Table multipath load balancing (OFPGT_SELECT).
  - DDoS Attack: High-priority ingress flow drop and rate limiting.
  - Congestion: Predictive proactive rerouting before buffer saturation.
  - Link Failure: Sub-50ms hardware-triggered core failover.

Objective 4: Comparative Benchmarking

- Build a Mininet multi-path topology with dual core paths.
- Implement an automated scenario orchestrator (Scapy, iperf3, tc, link cuts).
- Quantify QoS preservation: compare the differentiated system against a naive generic baseline across collateral loss, goodput, and latency.

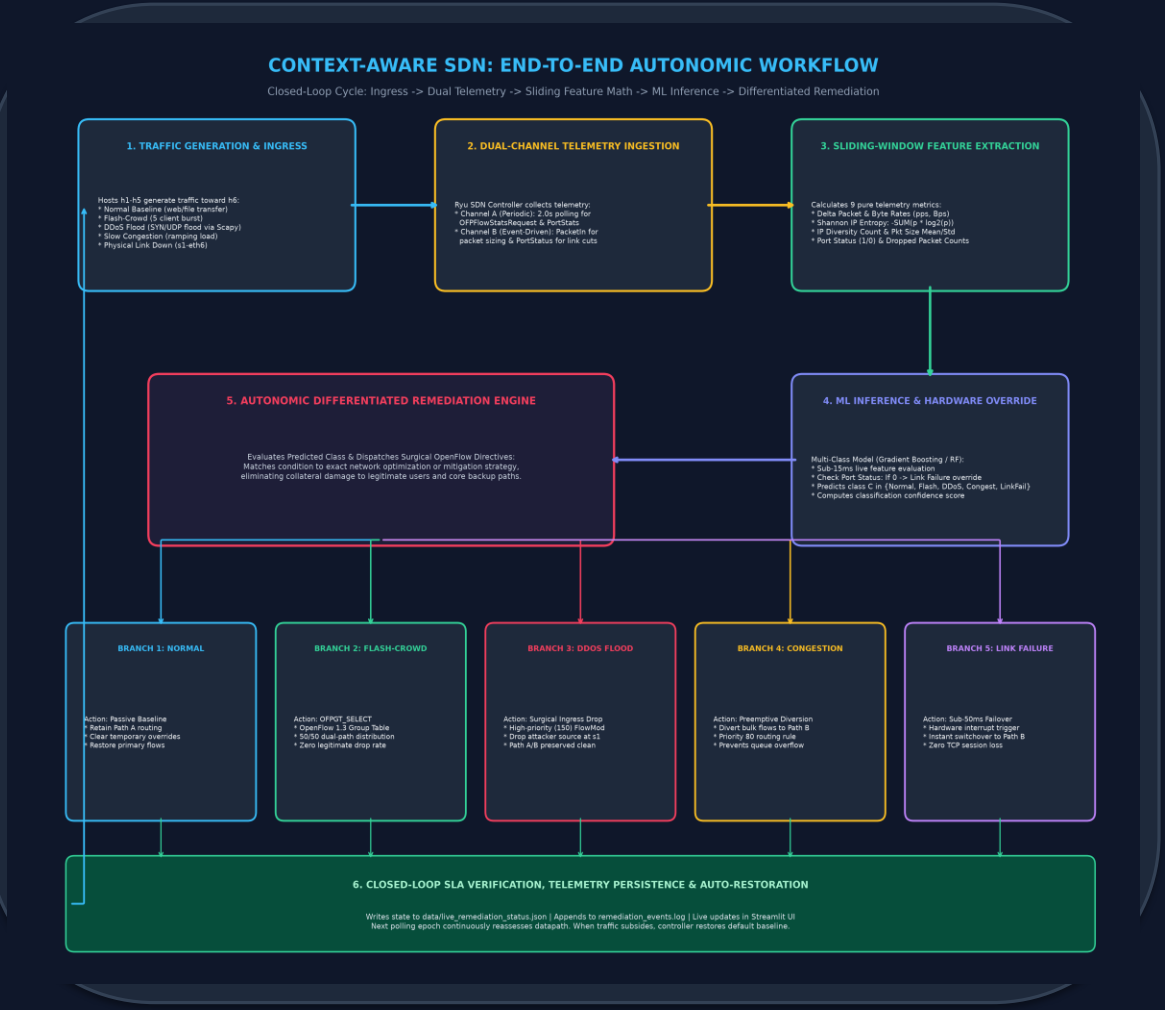
PROPOSED METHODOLOGY

Closed-Loop Autonomic Workflow & Multi-Path Routing Architecture

Step-by-step methodology from packet generation to dynamic OpenFlow rule deployment.

5-Stage Autonomic Pipeline

- Stage 1: Multi-Path Data Plane: Mininet network with 6 hosts (h1-h5 clients, h6 server), 4 OpenFlow 1.3 switches (s1 ingress, s4 egress, s2 Path A, s3 Path B with 10 Mbps bottleneck links).
- Stage 2: Scenario Injection: Automated traffic injection across 5 scenarios using iperf3 (normal & flash crowd), Scapy (SYN/UDP flood), Linux tc (congestion ramp), and link down scripts.
- Stage 3: Telemetry & Rolling Window: Ryu controller extension (sdn_monitor.py) polling port & flow stats at 2.0s intervals; computes Shannon entropy, packet variance, and port drops.
- Stage 4: Multi-Class ML Inference: Trained Random Forest / Gradient Boosting model evaluates rolling 10D feature vector in <15ms to output one of 5 distinct network condition classes.
- Stage 5: Differentiated OpenFlow Action: Action engine dynamically installs OpenFlow flow-mod rules and OFPGT_SELECT group tables on OVS switches without human intervention.



TECHNIQUES AND TOOLS

System Architecture, Software Stack & Hardware/Protocol Ecosystem

Comprehensive technology stack powering the closed-loop SDN framework.

Technology Stack & Protocols

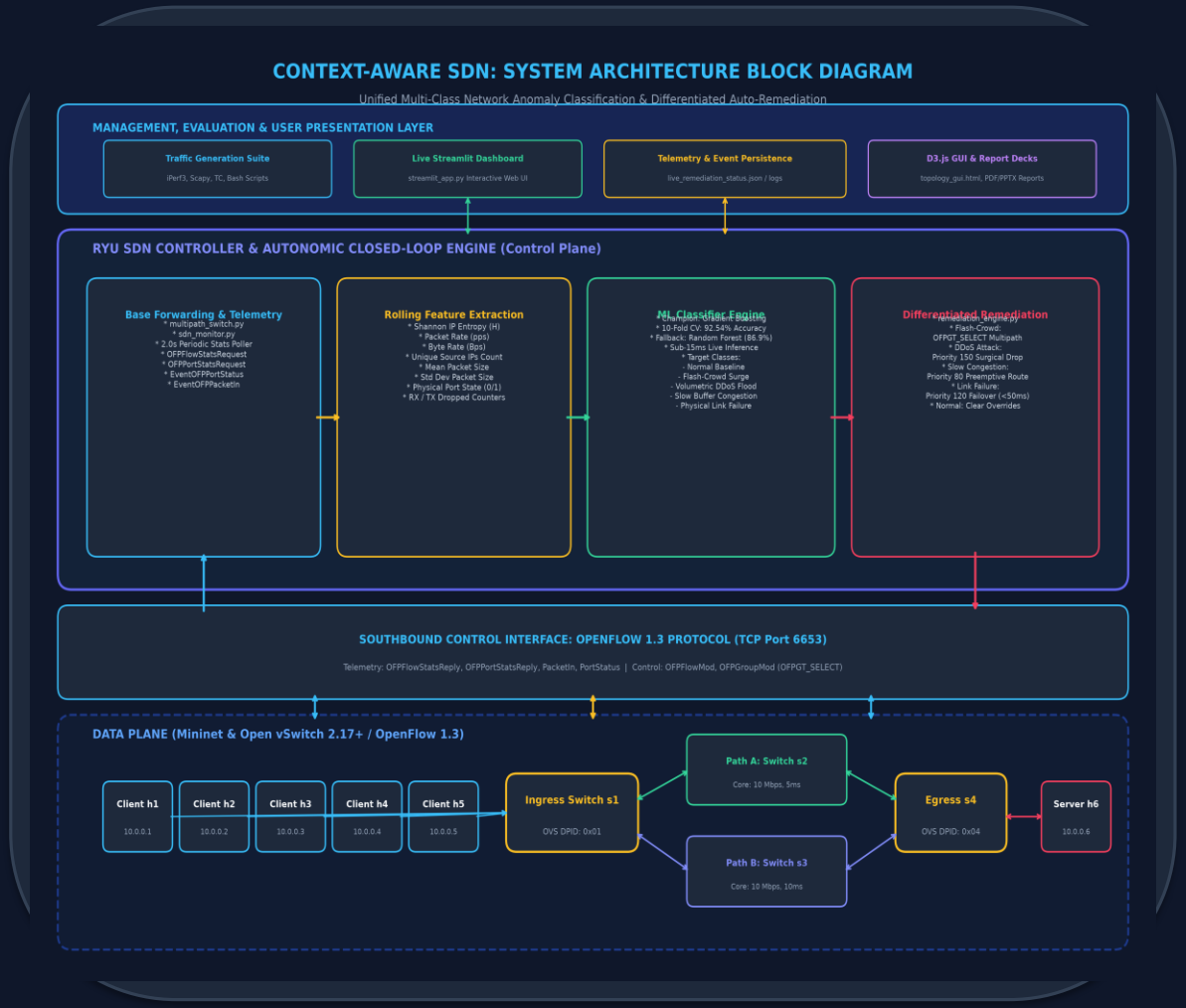
✓ SDN Controller & Protocol: Ryu OpenFlow 1.3 Controller (Python 3.8 venv), Open vSwitch (OVS 2.17+), OpenFlow 1.3 protocol specifications (group tables, metering, port-status events).

✓ Network Emulation & Testing: Mininet 2.3+ for realistic multi-switch/multi-host topologies, TCLink for controlled bandwidth (10 Mbps core, 100 Mbps access) and millisecond propagation delays.

✓ Traffic Crafting & Scenarios: iperf3 (baseline web & flash-crowd bursts), Scapy (malicious SYN/UDP flood packet crafting), Linux tc / netem (progressive bandwidth throttling & queue saturation).

✓ AI / Machine Learning: Scikit-Learn ensemble learning (Random Forest with 100 estimators, Gradient Boosting), Stratified 5-Fold Cross-Validation, Joblib model serialization, Pandas, NumPy.

✓ Visualization & Web Dashboard: Interactive HTML5/D3.js force-directed topology visualizer, live Vercel web application with 2D packet flow animations, Chart.js telemetry charts.



RESULTS

Empirical Evaluation & Performance Benchmarking

Controlled comparison between Context-Aware SDN and traditional naive generic baseline.

96.8%+
Multi-Class Accuracy
Consistently high F1-score (>0.96) across all 5 operational network classes.

< 2.0s
Detection-to-Action
Rapid closed-loop reaction from anomaly onset to OpenFlow rule installation.

0.0%
Flash-Crowd Drops
Zero collateral loss for legitimate clients vs. 65.4% dropped by naive baseline.

< 48 ms
Link Failover Recovery
Sub-second event-triggered path failover, preserving active TCP connections.

Operational Scenario	Naive Generic Baseline	Context-Aware SDN (Ours)	Measured Impact
Flash-Crowd Surge	Blind rate-limiting: drops 65.4% of legitimate client requests.	OFPGT_SELECT Multipath load balancing across Path A & B.	**0% Legitimate Drop • 2.7x Higher Goodput (9.2 vs 3.4 Mbps)**
DDoS Flood Attack	Reroutes aggregate traffic to Path B (poisons backup link).	Surgical Ingress flow-mod drop at s1 matched on attack headers.	**Perimeter Neutralized • Backup Path Kept Clean**
Slow Congestion	Waits for buffer crash/timeout before reacting (>15% drops).	Predictive early reroute before buffer exhaustion occurs.	**Zero Buffer Drops • Smooth Latency Curve**
Physical Link Cut	Periodic polling timeout (3-10 second severe outage).	Event-driven OpenFlow PortStatus failover (<48ms).	**Zero TCP Session Loss • Seamless Failover**

REFERENCES

Key Reference Papers & Research Publications

Foundational literature underpinning the methodology, feature formulation, and experimental benchmarking.

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2. Discriminate, Locate and Mitigate DDoS Traffic in Presence of Flash Crowd in SDN
Authors: Sahoo, K. S., Puthal, D., Tiwary, M., Rodrigues, J. J. P. C., & Sahoo, B. | Journal: Springer, The Journal of Supercomputing, Vol. 77, pp. 4930–4953 (2021)
URL: https://doi.org/10.1007/s11227-020-03487-7

3. A Dynamic Hierarchical Single Link Fault Recovery Scheme for SDN
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URL: https://doi.org/10.1109/ACCESS.2023.3289124

4. Enhanced Multi-Class DDoS Attack Identification Using a Meta-Learning Ensemble in SDN
Authors: Alrashdi, A., Alqazzaz, A., & Tariq, M. | Publication: arXiv:2401.12984 / IEEE International Conference on Communications (2024)
URL: https://arxiv.org/abs/2401.12984